Convex Optimization Exercises - Sheet 2

Exercise 5. Decide whether the following sets are convex:

(i) $\{(x, y) \in R_{++} \mid x/y \leq 1\}$

(ii) $\{(x, y) \in R_{++} \mid x/y \geq 1\}$

(iii) $\{(x, y) \in R_{++} \mid xy \leq 1\}$

(iv) $\{(x, y) \in R_{++} \mid xy \geq 1\}$

Exercise 6. Show that the set $\{x \in \mathbb{R}^n \mid Ax \leq b\}$ for given $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$ is indeed a convex set.

Exercise 7. Consider the linear program:

$$\begin{aligned} \text{minimize} \quad & f(x_1, x_2, x_3) = 2x_1 + 3x_2 - x_3 \\ \text{subject to} \quad & x_1 - x_2 \geq 4 \\ & 3x_1 + x_2 \leq 1 \\ & x_3 \leq 0 \end{aligned}$$

Find an equivalent linear program of the form:

$$\begin{aligned} \text{minimize} \quad & c^T y \\ \text{subject to} \quad & Ay = b \\ & y \succeq 0 \end{aligned}$$

(*Hint:* start by turning existing inequality constraints into an equality constraint with the help of an additional non-negative variable.)

Exercise 8. You are moving into a new apartment, your bedroom and the items there have the following measurements:

- **Room:** 3.5 m × 3.0 m
- **Rug:** 3.0 m × 2.5 m
- **Bed:** 2.0 m × 1.6 m
- **Wardrobe:** 1.0 m × 0.6 m
- **Desk:** 1.0 m × 0.5 m
- **Nightstand:** 0.4 m × 0.4 m

How to arrange the furniture, such that your beautiful rug is as exposed as possible? Visualize your solution. What constraints would you add to make the layout more practical?